batmonitor 0.65-c: one cycle (bm_step, every 5 s) | max persistent decision states 36864 (0.62-c: 3072) = D1 2 x D2 128 x lead 3 x D3 2 x D4 2 x D6 4 x unit 3 | blue = pure decision function (bits + state, enumerated by make check) | green = discharge matrix | [CBMC] = proved for all inputs

inputs (chain_inputs)
PCC, Sofar Bat1 W + SoC, em0 WP W, R290 Hz, PV,
BMS SoC / U / CCL / DCL / lowest cell (0.52-c)
-> stale, wp_on (em0 fresh and ≥ WP_ON_TH), night (PV < NIGHT_PV_TH)

D1 bm_d1 state: bat1_first, wp_low
B1FIRST: Bat1 SoC > BAT1_SOC_MIN (on from +2 %) and WP on
WPLOW: lowest stack < WP_SOC_MIN 21 % (back ≥ 23 %)
[make check: 192 combinations]

WPLOW

src

S0 gates per bank (chain_gates)
SoC from the BMS; caps per phase, one per bank:
chg_cap = min(charger x U, BMS CCL) dis_cap = min(DISCHARGE_MAX_PHASE, BMS DCL)

house = PCC + own charge (+ WP while WPLOW)
0.59-c: the real PCC, target 0 W; the heat pump is house load
(removed: Z2 winter rule, 1900 W transition cap, forecast rule, WPPV, WP_CAP)

S3 source (0.59-c: no season)
day -> **DAY**
night + B1FIRST -> **B1**
night, WPLOW + WP on -> **WPG** (grid pays the WP)
else night -> **STK** (night floor)

surplus

D2 bm_d2 per bank state: prot, force, low, chg_prev
first match: BMS_MISSING > FORCE (force or low) > STALE > LADESPERRE > CHARGE / FULL > PROT > CHARGING > **DIS**
LOW (0.52-c, state low): on at DCL < DCL_FORCE_A 1 A or lowest cell < CELL_FORCE_V 3.00 V, off when DCL is back, the cell ≥ CELL_FORCE_RELEASE_V 3.25 V and (0.55-c) SoC ≥ SOC_MIN_RELEASE; LOW -> FORCE (charge, no discharge); Lenggries env: CELL_FORCE_V 3.05 V
make check: 16128 combinations per stack (84 bit sets x 12 global x 16 states) -> mode:
BMS_MISSING 8064 · FORCE 5760 · STALE 1152 · LADESPERRE 288 · CHARGE 216 · PROT 216 · CHARGING 180 · DIS 180 · FULL 72
[CBMC h_d2_bits; threshold order needed: h_d2_bits_any_order]

row

house

grid_ok (vebus ActiveIn)

D-lead bm_dlead state: lead
|SoC1 - SoC2| > SOC_BALANCE_ON -> the stack ahead leads
< SOC_BALANCE_OFF -> no lead

mode per phase

else

DIS

CHARGE

BMS_MISSING / FORCE / STALE / LADESPERRE / FULL / PROT / CHARGING
-> fixed setpoint (0, force charge incl. LOW, ...)

charge (chain_charge)
want = max(0, own_charge + KP x surplus)
alloc_charge: CHARGE_PRIORITY, lead stack last, up to chg_cap
[CBMC h_charge]

dis_cap

lead

discharge matrix (0.45-c, whole watts)

BM_SRC_RULE[src]				
SRC	b1_signed	trickle	idle_w	prop_hold
DAY	0	0	20	0
B1	0	0	20	0
STK	1	1	0	1
WPG	0	0	20	0

deficit (int W)
b1 = b1_signed ? Bat1 : max(Bat1 charge, 0)
target = SOYO_TARGET + (trickle ? SOFAR_TRICKLE : 0)
deficit = own_discharge - b1 - KP_PM x (house - target) / 1000

D3 bm_d3 state: prop_prev
PROP = import (< -100 W) or
((prop_prev or prop_hold) and deficit > 20 W)

dis_amount: w = min(PROP ? max(deficit, 0) : idle_w, W_MAX)
[CBMC h_dis_amount]

D6 bm_d6 + S5a single_phase (0.63-c / 0.65-c): one MultiPlus while the grid is there [make check: 96 combinations (5 bits x 4 states), 14 one unit, 4 states reachable]

why: a MP2 loses ~30-40 W fixed + poor efficiency at low power; pv_victron 18.-26.10.2026:
discharge < 2 kW total lost 14 %, 2-8 kW 4-5 %, > 80 % of the minutes < 2 kW.
Grid there: the meter nets the phases -> one unit feeds, the others STBY (unbalanced on purpose).
Off-grid (emergency power): every phase carries its own load -> all units feed synchronously.

state single_all, single_over_since (W = SINGLE_DIS_W 3000, H = SINGLE_DIS_HYST_W 500, T = SINGLE_DIS_UP_S 60 s)				
grid_ok	**W > 0**	**amount w**	**next state**	**split**
0	-	-	unchanged	ALL (synchronous, off-grid)
1	0	-	unchanged	ALL (= 0.62-c)
1	1	w < W - H	ONE, over 0	1PH
1	1	W - H ≤ w ≤ W	kept, over 0	1PH if ONE, ALL if ALL
1	1	w > W, < T s	kept, over runs	1PH if ONE (up to dis_cap)
1	1	w > W, ≥ T s	ALL	ALL
1PH but w > dis_cap of that phase (3800 W / BMS DCL)				ALL this cycle

ONE single_all 0 over 0	-- w > W --> <-- w ≤ W --	**OVER** single_all 0 over_since set	-- T s -->	**ALL** single_all 1
ALL -- w < W - H --> ONE (ALL with W - H ≤ w stays ALL; ALL + over while w > W)				
invariants (make check): one unit only with grid + W on; off-grid all phases, state untouched; all phases only after T s above W; back only below W - H; fixed point after one step (no flapping)				

which unit (state single_ph)
lead stack (D-lead) -> its first phase
else the kept one while its SoC ≥ other - SOC_BALANCE_OFF
else the fuller stack (equal SoC: Stack1)
Stack1 -> L1, Stack2 -> L2 (first discharging phase)
rule tags: **1PH** (feeds w) / **STBY** (0 W)

ALL

1PH / STBY

alloc_discharge on **alloc_levels** (0.48-c: one split core for charge and discharge)
lead: level 0 = lead stack (BAL), level 1 = only what its caps cut (SPILL / WAIT)
no lead: one level, every stack the same share (EQ)
within a level: water filling up to dis_cap
[CBMC h_alloc_discharge: within caps, total ≤ w, lead first, equal per stack]

setpoints per phase -> Multis (ESS AcPowerSetpoint)
rule text: <mode> SRC:<src> <split: EQ/BAL/SPILL/WAIT/1PH/STBY> [CCL/DCL] [WPLOW]
[CBMC h_discharge: whole discharge path]

PI prototype (shadow, not armed)